WORKED SOLUTION

STEP II 2007, Question 1

Pure • Binomial Approximations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q1

The key idea

A binomial series converges quickly only when the small quantity really is small, so the whole question is about rewriting a surd as a simple rational multiple of $\sqrt{1+\text{something tiny}}$. To do that we look for a perfect square (or cube) that, multiplied by 3 or 6, lands just next to a power of 10. Then k is small compared with 100 (or 1000), and a handful of terms gives the accuracy we need.

Solution

Part (i)

With $x = \frac{k}{100}$, the binomial series for $(1+x)^{1/2}$ has coefficients

$$\binom{\frac{1}{2}}{1} = \frac{1}{2}, \qquad \binom{\frac{1}{2}}{2} = \frac{\frac{1}{2}\cdot(-\frac{1}{2})}{2!} = -\frac{1}{8}, \qquad \binom{\frac{1}{2}}{3} = \frac{\frac{1}{2}\cdot(-\frac{1}{2})\cdot(-\frac{3}{2})}{3!} = \frac{1}{16}.$$

Substituting $x = k/100$,

$$\left(1+\frac{k}{100}\right)^{1/2} \approx 1 + \frac{k}{200} - \frac{k^2}{80\,000} + \frac{k^3}{16\,000\,000}.$$

(a) With $k = 8$ the left-hand side is $\sqrt{1.08}$, and the point is that $108 = 3 \times 36$:

$$\sqrt{1.08} = \frac{\sqrt{108}}{10} = \frac{6\sqrt{3}}{10} = \frac{3\sqrt{3}}{5}, \qquad \text{so} \qquad \sqrt{3} = \frac{5}{3}\sqrt{1.08}.$$

The series gives

$$\sqrt{1.08} \approx 1 + 0.04 - 0.0008 + 0.000032 = 1.039232,$$

and therefore

$$\sqrt{3} \approx \frac{5}{3} \times 1.039232 = 1.7320533\ldots \approx 1.73205 \text{ to five decimal places.}$$

(b)

Ask yourself Which value of k makes $1+\frac{k}{100}$ equal to 6 times a nice square?

We want $100+k=6m^2$ for some integer m with $6m^2$ close to 100, so that k is small. Since $6\times 16=96$, take $m=4$ and $k=-4$:

$$\sqrt{0.96}=\frac{\sqrt{96}}{10}=\frac{4\sqrt{6}}{10}=\frac{2\sqrt{6}}{5},\qquad\text{so}\qquad\sqrt{6}=\frac{5}{2}\sqrt{0.96}.$$

With $k=-4$ every term after the first is negative:

$$\sqrt{0.96}\approx 1-0.02-0.0002-0.000004=0.979796,$$

and so

$$\sqrt{6}\approx\frac{5}{2}\times 0.979796=2.44949 \text{ to five decimal places.}$$

This k is half the size of the one in (a), so the neglected terms are even smaller here. (Something like $k=50$, from $1.5=6/4$, also gives $\sqrt{6}$ exactly, but with $x=\frac{1}{2}$ the series converges far too slowly for five decimal places.)

Part (ii)

The first two terms give

$$\left(1+\frac{k}{1000}\right)^{1/3}\approx 1+\frac{k}{3000}.$$

Ask yourself Which cube, multiplied by 3, is just above 1000?

We need $1000+k=3m^3$ with k small. The cubes near $1000/3\approx 333$ are $6^3=216$ and $7^3=343$, and $3\times 343=1029$ is very close to 1000. So take $k=29$:

$$\sqrt[3]{1.029}=\frac{\sqrt[3]{3\times 7^3}}{10}=\frac{7\sqrt[3]{3}}{10},\qquad\text{so}\qquad\sqrt[3]{3}=\frac{10}{7}\sqrt[3]{1.029}.$$

Using the two-term approximation,

$$\sqrt[3]{3}\approx\frac{10}{7}\left(1+\frac{29}{3000}\right)=\frac{10}{7}\cdot\frac{3029}{3000}=\frac{3029}{2100},$$

which is the required approximation.

Checks

A calculator gives $\sqrt{3}=1.7320508\ldots$ and $\sqrt{6}=2.4494897\ldots$, which agree with both answers to five decimal places. For part (ii), $3029/2100=1.44238\ldots$ while $\sqrt[3]{3}=1.44225\ldots$. The error of about 1.3×10^{-4} matches the size of the first neglected term, $\frac{10}{7}\cdot\frac{1}{9}\left(\frac{29}{1000}\right)^2\approx 1.3\times 10^{-4}$, so the approximation is a slight overestimate.

Common slips

- Getting the third coefficient wrong: $\frac{1}{2}\cdot(-\frac{1}{2})\cdot(-\frac{3}{2})/6 = +\frac{1}{16}$, not $-\frac{1}{16}$.
- Forgetting to multiply back by $\frac{5}{3}$, $\frac{5}{2}$ or $\frac{10}{7}$ after evaluating the series.
- Choosing a large k (such as $k = 50$ in (b)), for which three or four terms are nowhere near enough.

Beyond STEP

How many terms you need comes down to bounding the remainder of a Taylor series. Lagrange's form of the remainder lets you prove, rather than just believe, that these approximations are correct to five places. The same trick of rescaling into the region where a series converges quickly is how calculators and computer libraries actually compute roots and logarithms. For the theory of Taylor series and their remainders, see Abbott, *Understanding Analysis*.

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